EST I Literacy & Quantitative Diagnostic Drill

PRACTICE WORKSHEET

No-Calculator & Calculator Readiness • Official EST I Curriculum

Name: _____ Date: _____ Score: ____ / 25

EST I Literacy & Quantitative Diagnostic Drill Practice Exercises

1 MCQ

Consider the functions $f(x) = 2x + 3$ and $g(x) = x + 7$. At what x-value do the graphs of $f(x)$ and $g(x)$ intersect?

A 4

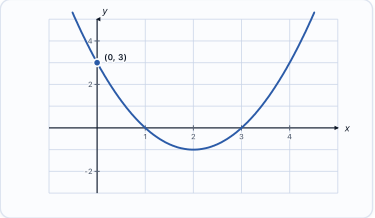
B 7

C 3

D 10

2 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(0)$?



A -3

B -1

C 1

D 3

3 MCQ

In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle $\angle AGH$ is 125°, what is the measure of angle $\angle GHD$?

Diagram description: Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G



A 55°

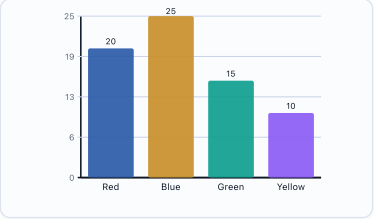
B 125°

C 90°

D 180°

4 MCQ

The bar chart below shows the number of students who prefer each of four different colors. What is the total number of students who prefer either blue or green?



A 30

B 40

C 50

D 60

5 MCQ

A recipe calls for 2 cups of flour for every 3 cups of sugar. If 8 cups of flour are used, how many cups of sugar are needed?

A 10 cups

B 12 cups

C 11 cups

D 13 cups

6 MCQ

If $9^{x+1} = 27^{x-1}$, what is the value of x?

A 2

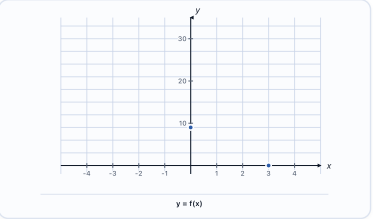
B 5

C 3

D 7

7 MCQ

The graph of the quadratic function $y = x^2 - 6x + 9$ is a parabola. What is the x-coordinate of the vertex of this parabola?



A 0

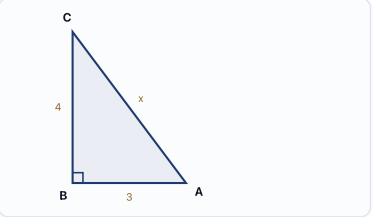
B 6

C 3

D -3

8 MCQ

In the right-angled triangle ABC shown below, $\angle ABC = 90^\circ$, $AB = 3$, and $BC = 4$. What is the length of the hypotenuse AC?



A 5

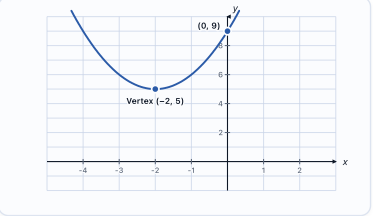
B 7

C 12

D 25

9 MCQ

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at the point $(-2, 5)$ and the parabola passes through the point $(0, 9)$. What is the value of b?



A -2

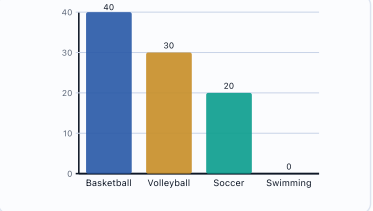
B 2

C -4

D 4

10 MCQ

The bar chart below shows the number of students who participated in different sports. The number of students who participated in swimming is unknown and represented by x. If the total number of students who participated in all sports is 120, and the number of students who participated in basketball, volleyball, and soccer are 40, 30, and 20 respectively, what is the value of x?



A 30

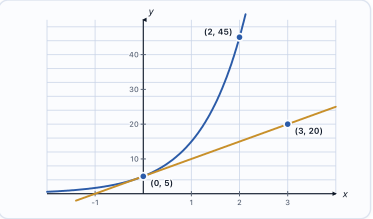
B 20

C 40

D 50

11 MCQ

The diagram shows the graphs of the exponential function $y = a \cdot b^x$ and the linear function $y = mx + c$. The exponential function passes through the points $(0, 5)$ and $(2, 45)$. The linear function passes through the points $(0, 5)$ and $(3, 20)$. Find the value of a.



A 5

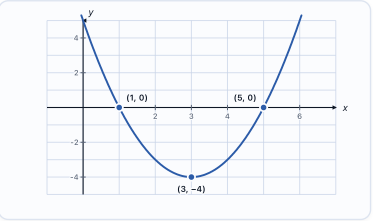
B 3

C 9

D 1

12 MCQ

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?



A 1

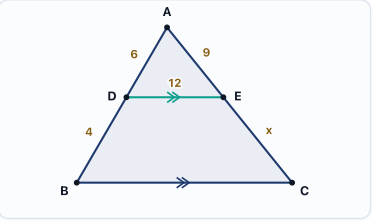
B -1

C 2

D -2

13 MCQ

In the diagram below, line segment DE is parallel to line segment BC. If $AD = 6$, $DB = 4$, and $AE = 9$, what is the length of EC?



A 7.2

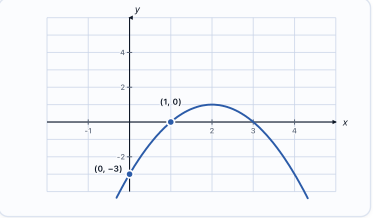
B 10

C 6

D 4.5

14 MCQ

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.



A 3

B 4

C 5

D 6

15 MCQ

If $i = \sqrt{-1}$, what is the value of i^{197}?

A i

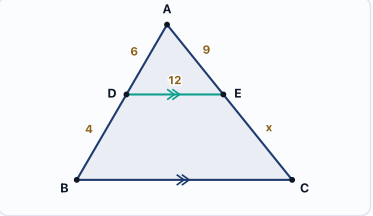
B 1

C -1

D $-i$

16 MCQ

The diagram shows a triangle ABC, where D is a point on side AB and E is a point on side AC. It is given that DE is parallel to BC. If AD = 6 cm, DB = 4 cm, AE = 9 cm, and DE = 12 cm, find the length of the side BC.



A 24 cm

B 30 cm

C 20 cm

D 18 cm

17 MCQ

A technician charges a flat initial fee of $62 plus an hourly rate of $21 for each hour of labor on a service call. The total charge C in dollars for h hours of work is given by $C = 21h + 62$. What is the best interpretation of the number 21 in this context?

A The number of hours needed to complete the service call

B The total cost for 62 hours of service

C The minimum flat fee charged for the visit, $21

D The hourly labor rate of $21 per hour

18 MCQ

A theater sells adult tickets for $\$14$ and child tickets for $\$10$. On a night when 12 tickets were sold for a total of $\$148$, how many adult tickets were sold?

A 7

B 6

C 8

D 5

19 MCQ

Data Set A has values clustered closely around its mean of 50, ranging from 48 to 52. Data Set B has the same mean of 50, but its values are widely dispersed, ranging from 20 to 80. Both sets have 100 observations. Which statement correctly compares their standard deviations, s_A and s_B?

A There is not enough information to compare the standard deviations without individual data values

B $s_A > s_B$, because Data Set A is more densely concentrated

C $s_A < s_B$, because Data Set B has greater variability around the mean

D $s_A = s_B$, because both sets have identical means of 50

20 MCQ

Consider the function $f(x)$ defined by $f(x) = |x - 2| + |x + 1|$. Find the minimum value of $f(x)$ and the interval over which this minimum is attained.

A Minimum value 2 on the interval $[-1, 2]$

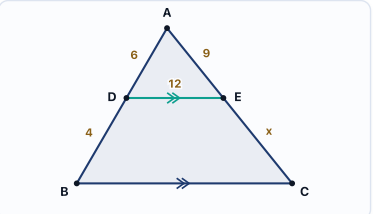
B Minimum value 3 on the interval $(-\infty, -1]$

C Minimum value 3 on the interval $[-1, 2]$

D Minimum value 2 on the interval $[2, \infty)$

21 MCQ

In the given diagram, $\triangle ABC$ is inscribed in a circle. Points D and E lie on sides AB and AC respectively. It is given that $AD = 6$, $DB = 4$, $AE = 9$, and DE is parallel to BC. Furthermore, $DE = 12$. Calculate the length of BC.



A 24

B 20

C 27

D 18

22 MCQ

Which of the following is equivalent to the expression $\frac{x^2-10x+24}{x^2-36}$ for all $x \neq \pm 6$?

A $\frac{x-4}{x+6}$

B $\frac{-10x+24}{-36}$

C $\frac{x+4}{x-6}$

D $\frac{x-4}{x-6}$

23 MCQ

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score:

The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 5x + 60$.

Hours Studied (x)	Exam Score (y)
2	65
3	70
4	78
5	82
6	88
7	91
8	95
9	98
10	100
11	103

A 92.5

B 95.0

C 97.5

D 100.0

24 MCQ

Which of the following is equivalent to $\log_b\left(\frac{x^5y^3}{z^4}\right)$ for all positive values of x, y, z?

A $5\log_b(x) \cdot 3\log_b(y) - 4\log_b(z)$

B $\frac{5\log_b(x)+3\log_b(y)}{4\log_b(z)}$

C $5\log_b(x) - 3\log_b(y) - 4\log_b(z)$

D $5\log_b(x) + 3\log_b(y) - 4\log_b(z)$

25 MCQ

How many different integers between 15 and 45 contain only digits from the following list: 1, 3, 5, 7, 9 ?

A 16

B 6

C 5

D 7